

Aditya Wadhwa

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Education

B.Tech in Computer Science and Engineering

University of Petroleum and Energy Studies (UPES), Dehradun

Higher Secondary Examination (12th Grade)

Manav Rachna International School, Faridabad, Haryana

Senior Secondary Examination (10th Grade)

Manav Rachna International School, Faridabad, Haryana

Aug 2023 – Present

CGPA: 8.66

Apr 2022 – Mar 2023

Percentage: 92.8%

Apr 2020 – Mar 2021

Percentage: 92.2%

Internships

Orchid ENT Pvt Ltd — Toscee Collective

IT & Automation Intern

June 2026 – August 2026

Noida, India

- Analysts were spending **4+ hours manually** parsing purchase orders — designed and shipped an LLM pipeline (scraping → extraction → cross-combination recommendations) that cut this to **under 15 minutes**.
- Business reports were locked in siloed PDFs with no queryable interface — built an **MCP integration with Claude** that exposed structured data as reusable AI tools, letting non-technical stakeholders **query reports in natural language**
- Translated ambiguous operational requirements into practical automation workflows covering **data extraction, transformation, analysis, and AI-assisted decision making**.

Projects

Multimodal Multi-Agent RAG Framework | Python, FAISS, GraphRAG, LLMs, HPC

- Existing RAG systems fail on multi-hop questions across **text, tables, and figures** — built a **five-agent pipeline** that reasons across all three modalities end-to-end.
- After profiling retrieval failures, introduced **MonoVLM reranking and graph traversal** on top of FAISS — improving **RAGAS metrics faithfulness to 0.94, context precision to 0.89, and answer relevancy to 0.88**.
- Developed GPU-optimized embedding workflows with hybrid **FAISS and Knowledge Graph indexing** for scalable document processing on HPC clusters.

MyLegalSahayak (AI Legal Tech Startup) | Node.js, Gemini API, MongoDB, RAG

- Legal advice in India is inaccessible to most — built a multilingual AI legal assistant grounded on **400+ Indian legal documents**, using **RAG and cosine-similarity retrieval** to ensure cited, factual responses.
- Raw LLM responses were drifting off-topic and generating **unsupported legal claims** — engineered a **relevance guardrail** that blocks off-topic queries before generation, **reducing hallucination risk** in a high-stakes domain.
- Engineered a document-analysis module capable of processing complex PDFs up to **25MB**, cross-referencing an Indian laws database, and generating structured risk reports with clause-level alerts.
- Designed a unified **vector knowledge base of 400+ legal documents** using cosine-similarity retrieval to inject relevant, grounded context into every LLM generation.

Core Competencies

Languages: Python, Java, JavaScript, SQL

Backend & Web: Node.js, Express.js, React.js, REST APIs

AI & LLM: LLMs, VLMs, RAG Pipelines, Multi-Agent Systems, Prompt Engineering, Context Engineering, NLP, AI Guardrails

AI Tools: Claude, Gemini, ChatGPT, Model Context Protocol (MCP), AI-Assisted Development

ML & Retrieval: TensorFlow, PyTorch, Scikit-learn, FAISS, BERTopic, Vector Search

Databases: MongoDB, Knowledge Graphs

Achievements & Co-Curricular

Nyayniti Associates | Legal Tech Intern

2025

- Analyzed legal document workflows and gained domain expertise, directly informing the architecture and relevance guardrails for the MyLegalSahayak AI startup.

Ten Smiles Foundation (NGO) | Volunteer Educator

Faridabad, India

- Taught foundational computer skills to educators of underprivileged children, leveraging strong technical communication to explain complex concepts to non-technical users.